Sixdof Space
ULTRA FAST OPTICAL TRACKING

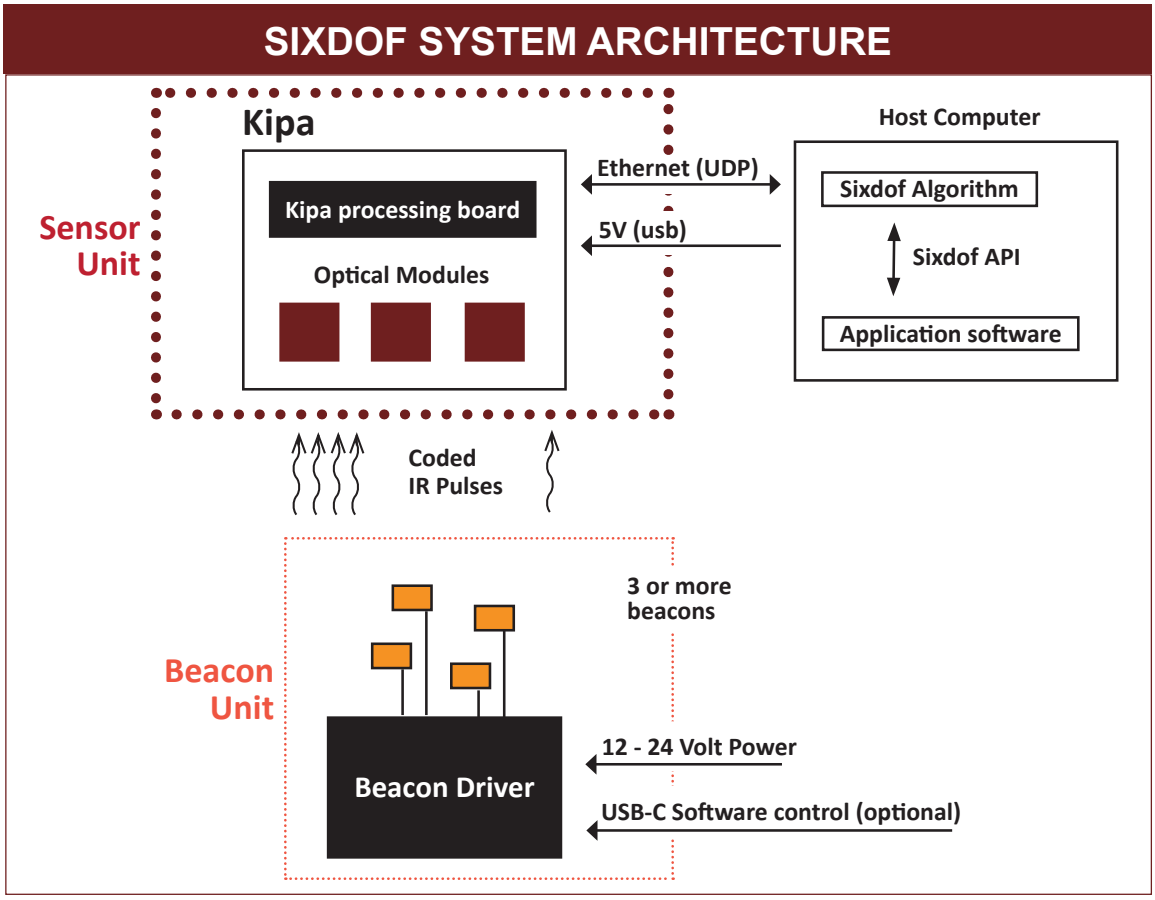
Kipa Sensor Data Sheet

The Kipa and Kipa Pro sensor units provide equivalent core capabilities and operate on a unified software platform, enabling effortless interchangeability within the product line. The Kipa Pro is NDAA compliant. Mechanical form factor, including the dimensions and location of the connector, differ between the two products (see drawings, page 3).

HIGHLIGHTS

- High-Resolution Drift-Free Optical Tracking
- Fully independent of GPS
- Operates under conditions of total darkness to full sunlight
- High toleration for strong winds and adverse weather conditions
- Full 6DOF Pose (position and orientation) with cm-level accuracy
- Tracking up to 100 meters
- Includes simple-to-use SDK with C++ API, Python wrapper, compatible with both ROS2 and DDS
- Simple installation and maintenance
- High Speed (updates every 2.5 ms)
- Compatible with Windows, Linux, Android, and Raspberry PI operating systems

SIXDOF SYSTEM ARCHITECTURE



BENEFITS

- Precision low-latency angular and 6DOF tracking
- Low computational requirements (no DSP/GPU)
- Continuous optical lock during high-rate movement
- Clearly defined API enables a range of use cases
- Software includes system integration examples

PERFORMANCE

Absolute Angular Accuracy
<0.5° over 120° FoV

Absolute Positional Accuracy
As low as 1 cm
(Dependent on distance and beacon baseline.)

Field-of-view
120° x 120°

Operating light levels
0 – 100,000 lux

Interface
Ethernet (UDP)

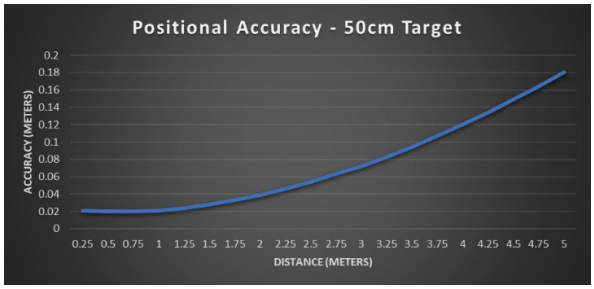
Power supply (sensor)
5V, 250mA

Operating temperature
-30°C to 50°C

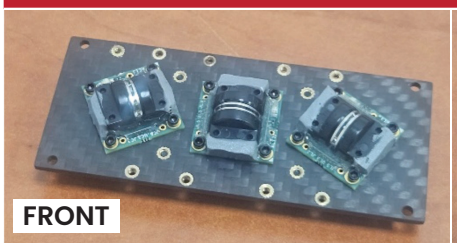
Raw measurement frequency
Over 10,000 FPS

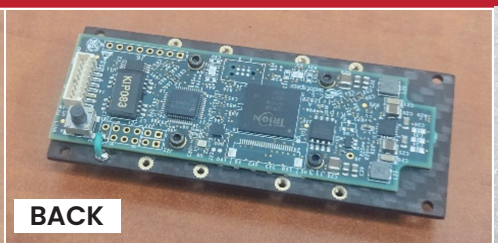
Capture latency
<2ms

Host processor requirement
1 x 2GHz core



Kipa/Kipa Pro sensor board mounted on a carbon plate





Size (mm): 92 x 38 x 18
Weight: 27 g



Size (mm): 98 x 44 x 20
Weight: 62 gr with plastic filter
75 gr with glass filter

PART NUMBERS

010-1001-000
Kipa Sensor - No Housing (carbon plate, 8Pin Back)

010-1001-001
Kipa Sensor - in Housing with Plastic Filter (8Pin Back)

010-1001-002
Kipa Sensor - in Housing with Glass Filter (8Pin Back)

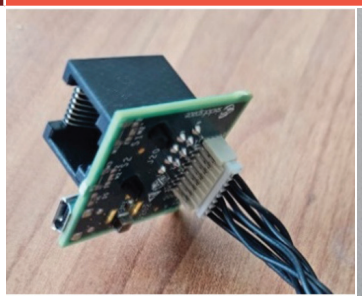
010-1001-003
Kipa Sensor - No Housing (carbon plate, 5Pin Back)

010-1006-000
Kipa Pro Sensor - No Housing (carbon plate, 8Pin Side)

010-1006-001
Kipa Pro Sensor - in Housing with Plastic Filter (8Pin Side)

010-1006-002
Kipa Pro Sensor - in Housing with Glass Filter (8Pin Side)

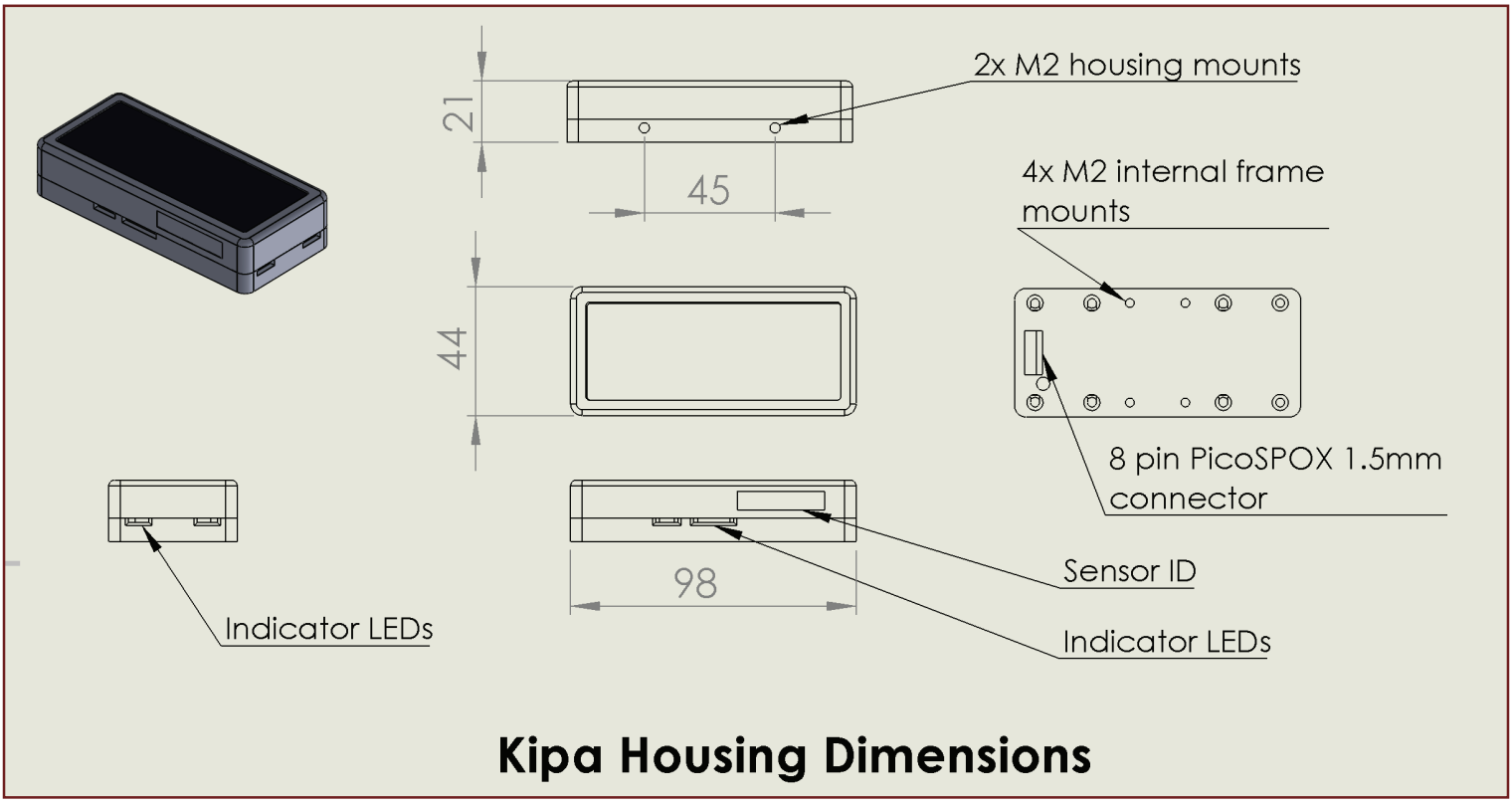
INTERFACE OPTIONS



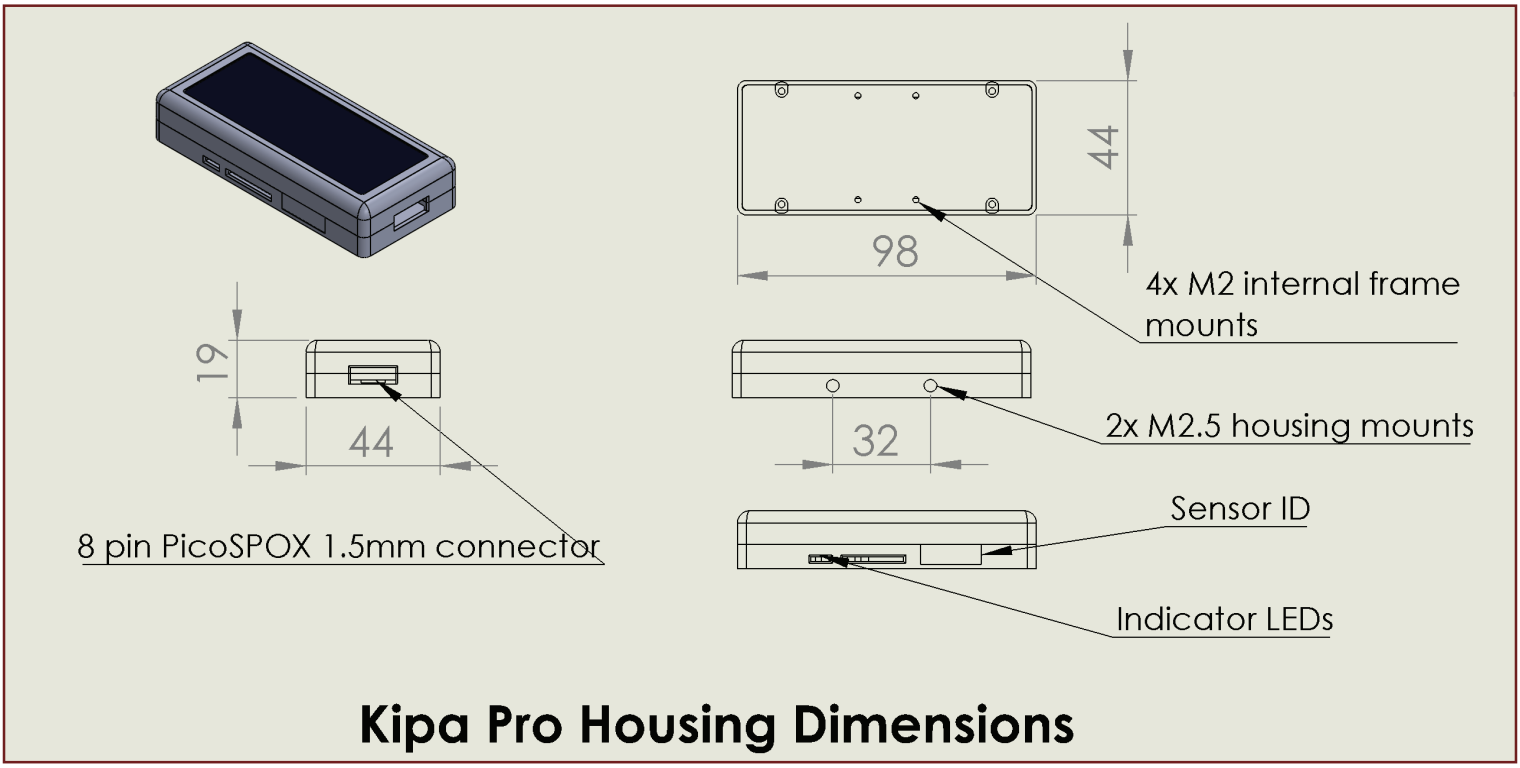
Size (mm): 30 x 27 x 20
Weight: 6 g
Network: RJ45
Power: USB
Molex 8pin Pico-SPOX connector, 10 cm cable
Part #:040-1002-000



Molex 8pin Pico-SPOX connector to RJ45 and USB (power).
Part #:040-1001-000 (1m)
Part #:040-1001-001 (3m)



Kipa Housing Dimensions



Kipa Pro Housing Dimensions

Elevate your tracking and positioning technology with Sixdof's unmatched optical tracking precision to enable autonomous robotic and drone control.

For more details, visit us at www.sixdofspace.com

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